

30 Storytelling and Sensemaking through Data Visualization

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A picture is worth a thousand words.

30.1 What is Data Visualization?

Data visualization is a technique of distilling and communicating the value of data using a visual medium: it makes answers to complicated questions available in a digestible and engaging manner, while furnishing insights to better facilitate and guide leaders' decision-making about their workforce. A simple question guides our scientist-practitioner advice and approach to data visualization: Are we providing decision-makers with valuable insights for business action? By this, we expect several touchstones are achieved: (1) we are harnessing available data and consulting around potential new data sources; (2) we are finding ways to blend seemingly disparate data sources, thereby opening new vistas for exploration and discovery; and (3) we are leveraging data to facilitate more profound and insightful conclusions using descriptive, diagnostic, and prescriptive analytics to develop compelling stories that guide decision-making. By and large, data visualization has the potential to amplify human cognition – the mental structures and processes of attention, perception, analysis, communication, and memory – for large quantities of information.

From Neanderthal cave drawings of successful hunts to the pilgrimage of the Chicago Cubs ending their 108-year World Series drought, data visualization is not a new concept, though trends and discussions in the marketplace might leave you thinking otherwise. What has evolved, is increased attention paid to the “Vs” of data: volume, variety, velocity, veracity, value, and visualization (see De Mauro, Greco, & Grimaldi, 2016). Technological advances have made visualization available to the masses (e.g., Tableau, Domo, Microsoft Power BI, Qlik, Plotly, Sisense BI), though by no means easing the burden of knowing the right questions, structures, and kinds of approaches to data in order to fully visualize it and empower leaders. In fact, the bar is set higher to use the best methods and approaches to meet organizational needs, because the potential impact of decisions from data displays and analytics are far-reaching.

Data visualization sits on the forefront of many areas that pose challenges for those consulting and conducting analyses for organizations and leaders. In the

technological age, our patience for information has waned and the reality is that we all want information available at our fingertips. Because of technology, it is more possible than ever to provide data on a more immediate basis. Even so, all of the issues that confront data, such as methods, collection, structure, and analytics, are not eased by cutting-edge data displays and means for blending data together quickly. As the adage goes, “garbage in, garbage out,” and this perhaps makes data science and approaches even more challenging as data become easier to quickly grasp from a compelling image, but underlying issues due to structure or analytics become less visible, especially to untrained eyes. As an expert, it is incumbent to address underlying data issues, ensure data structures are correct, and ensure the analyses underpinning data displays remain appropriate, valid, and meaningful – both statistically and practically.

We make the following suggestions regarding research and data visualizations. Ensure all of the basic data essentials are met, which makes completing data visualization contingent on the following: having all of the right data that are necessary, that the data are accurate, and that the data are structured to effectively analyze questions. In short, having everything you need to accurately and effectively answer key questions from stakeholders, even anticipating other questions or information that will arise from the data analysis and presentation. The core of research and proper analytic techniques are always core to data visualization. A major component of good data visualization is value. While “value added” seems to be understood, it is perhaps one of the most overlooked features because it is too easy to focus on the research questions, data, insights, and whether or not the questions were sufficiently answered. As such, the research and insights from the data need to do more than simply answer the questions. They need to be compelling and meaningful, so that it continuously serves as a resource for the decision-makers leverages what was uncovered from the research.

When building data visualizations, we encourage the use and consideration of the visual analytic framework (Keim et al., 2008). The visual analytic framework goal is “to turn information overload into an opportunity” through the combined use of technology and human intelligence while addressing several components of data, analytics, and the presentation of both (p. 155). The visual analytic framework is quite useful, though we propose additional dimensions to enhance the presentation of information and the retention of it, which are design thinking, aperture, and control. The objective of design thinking applied to data visualizations is to create more effective visualizations that stimulate a step-wise, deeply considered approach to thinking outside the box to address complicated stakeholder concerns. Indeed, with high-level stakeholders there is an incredibly limited margin for error – you have one shot to convince – and a windfall opportunity if done right, to change the way something is perceived and approached within an organization. In essence, the presentation uncovers several insights as easily and directly as possible, focuses leaders, and allows them the ability to make maximally informed decisions. Just as an aperture changes the depth of field to create an intended image, so too should an aperture be applied to the abstraction of data. While there are many paths to guiding discovery with data visualization, many subscribe to

Shneiderman's (1996) visual information-seeking mantra: "overview first, zoom and filter, then details on demand" (p. 336). Furthermore, like any good presentation or piece of art, once it is created it is out in the open and subject to a myriad of users, interpretations, and uses, some of which exceed or misrepresent the original intent. The reality is that, at some point, the data visualization will no longer be in the control of the creator or presenter. Rather than being approached as a problem, the lack of control is an opportunity. As such, we leverage interactive control to ensure that once the visualization is created, the end users can make the discoveries that we wanted them to uncover, but we turn it over to them to make additional discoveries, engage with it, as well as minimize the opportunity for automation bias (Skitka, Mosier, & Burdick, 2000). We treat end users like consumers, hopefully creating a sense of the IKEA effect within the end user. Indeed, there is a substantial "increase in valuation of self-made products" that can even contend with those created by experts (Norton, Mochon, & Ariely, 2012, p. 453).

Think of what we are suggesting with interactive control as a balance of control and freedom. Most people can think of a backseat driver. Why do people insist on others driving more like themselves or want others to drive the way they drive? Data visualization will inevitably meet the same issue, where the audience wants to control the information and use it in their own way. Interactive control plans for this eventuality and sets up confines so that leaders can take and use the information, slice and evaluate it as they deem fit and want, but all of the underlying data analytics and work are done so the data are still accurate, representative, and useful. The design is to enhance engagement by expanding the realm of control available to the end user, in our case, the client.

Rather than be beleaguered by the complexities of presentational features, and focusing on the challenges, let's turn the issue on its head and think of the opportunities presented and freedom granted inside data displays to meet others' needs. A goal of data visualization is to make it a great experience, so the audience is more likely to understand and recall it. How can we create compelling and engaging presentations that answer questions leaders pose about their workforce and that raise awareness regarding issues about which they were previously unaware? Rather than trying to train yourself in everything, with whom can you partner and blend your skills, knowledge, and talents? What software can you use to leverage you toward your collective success? How can you make your presentations engaging and, even better, interactive so that you can meet the unique needs of those to whom you are presenting, while answering questions in a more fluid manner? Beyond other mentioned factors we strongly rely on and use, we use humble inquiry, active listening, parsimony, teamwork, and drawing on others' insights, especially those who have backgrounds different from our own, to make a presentation stronger. We also use technologies that facilitate our ability to address many organizational questions and issues, and to do so in an intuitive, proactive way. We demand that our data science approaches have the same high rigor when we distill and display those data insights. We want people to appreciate as many features of our presentations as possible, meanwhile making the presentation more interactive and engaging, captivating their focus and allowing them to

better attend to and retain the information. The goal is to move people through displays of information to action.

30.2 People's Perceptions Drive Perceptions of Results

Effective data presentations hinge on not only data components but also people. Coupled with the technological advances, data visualizations face challenges from biased information processing, selective attention, and several other factors that drive people's thoughts, attention, and behaviors. Like the visual analytic framework suggests, due to differences in human cognition and perception, it is important to pay attention to some key factors in variations in attention. People are selectively attentive, through their own volition or not, in learning new information or generally processing information (Deng & Sloutsky, 2016; Hoffman & Rehder, 2010). Studies in human cognition and learning show that, while people employ many strategies to process and encode new information, people generally optimize their processing of information through categorizing information and making inferences from it (Deng & Sloutsky, 2016). Generally, people tend to categorize and recall what they are exposed to via color, shapes, textures, and a variety of other dimensions. As such, color and shapes are an important part of data visualization, because they allow people to attach meaning and categorize the information that you are conveying and want them to remember. Colors and shapes are crucial to simplifying and clarifying the presentation of your story, but the information has to be salient and presented in a meaningful way, directly addressing concerns or interests. As a result, use images or colors to facilitate and ease mental processing. For example, presenting performance as "gold, silver, or bronze" may use heuristics and shapes to more readily attribute meaning to visuals.

There are incredibly complex mechanisms underlying the way people process information, and keep in mind that the human brain is wired for seamlessly processing information (Gazzaniga & Mangun, 2014). Human perception and attention is drawn to and influenced by movement, color, patterns, and many other features that are crucial to the way people process information. This means that these features of a presentation have the power to embolden a message or to mute it. A single color, for instance, has the potential to undercut a presentation if not carefully considered. The color red has different meanings in different contexts, and changes people's processing considerably. Red in social contexts is linked to people's perception of social dominance or aggression, such as a face flushing red (Elliot, 2015). In other contexts, like prior to performing a taxing cognitive task, red may undermine performance on the task, but red also can be beneficial in rallying team members together such as before sporting competitions (Elliot, 2015). This conversation about the color red is to make us all seriously consider the importance of color and design. The choices of colors and hues are typically considered after the fact, though they clearly will determine success in displaying information in a compelling manner. One needs to be mindful of the context in which one's displays are presented. The example of the various implications of using red also

offers an important reality check. Each one of the authors of this chapter is guilty of not fully measuring the importance of color contained in a presentation and has missed the mark as a consequence. Color, shapes, and other important features present challenges. In particular, they present us with an interdisciplinary challenge, especially because data gurus are unlikely to also be trained in human differences in attention and learning patterns, though they are expected to be able to meet people's diverse needs when presenting data. From our lessons learned, we recommend immediately starting with pure black and white designs. Make certain that your intended message is clear from these alone; if it is not, then color will not clarify an already muddled message.

30.3 People Data to Solve Business Problems: An Example

Fundamentally, human resources' (HR) goal is to maximize performance. As a result, HR continuously tracks and measures a variety of work quality, quantity, and efficiency metrics: turnover rate, absenteeism rate, time to fill, performance reviews, engagement, salary, promotion, training hours, sales, revenue per full-time equivalent, among countless others. Often, these metrics are blended, augmented, and analyzed to support decision-making – for example, succession planning, a process that evaluates an employee's current performance against the employee's potential for success in higher roles to be used for employment decisions. To demonstrate the potential power of data visualization, we created a fictitious succession planning data visualization given the following scenario:

Kokopelli Bank is a family-owned, local bank that takes a personal approach to providing a breadth of services across its twenty-six branches in Albuquerque and its surrounding area. After the District Manager's retirement announcement, Kokopelli Bank is faced with the dilemma of finding his successor. With change comes opportunity, and Kokopelli Bank is committed to developing and promoting talent from within. The Executive Committee knows that the industry is facing a difficult and uncertain period with increased regulations, and wants to leverage relevant data to guide their decisions when considering each of the twenty-six Branch Managers for promotion. Relevant data included the following:

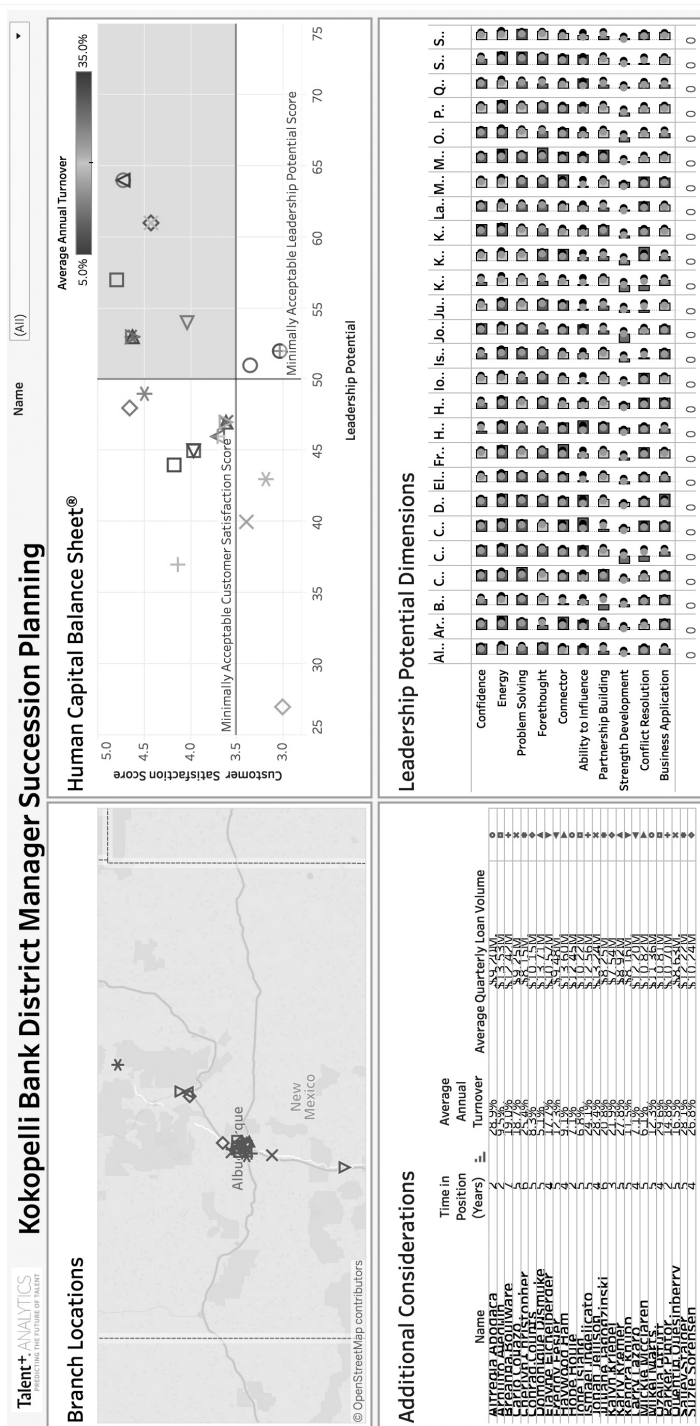
1. Branch Manager Demographics: Name, tenure, and branch location.
2. Branch Manager Performance: Branch customer satisfaction score (identified as the most important indicator of a Branch Manager's performance by the Executive Committee), average annual turnover rate, and average quarterly loan volume.
3. Branch Manager Leadership Assessment: Validated, personal characteristics that measure leadership potential and predict future success. The assessment was administered during the pre-employment application phase to be considered for Branch or District Manager positions.

Our team acted as the external Talent Analytics Consultant, hired by the Executive Committee to harness, aggregate, and augment these data into a meaningful data

visualization. Figure 30.1 presents our conceptualization of a dashboard solution that informs promotion decisions and succession planning strategies. We strongly recommend you interact with this dashboard for yourself, as if you were an Executive Committee member, by visiting: <https://public.tableau.com/profile/benjamin.sievert#!/vizhome/CambridgeHandbookofTechnologyandEmployeeBehaviorStorytellingandSensemakingwithDataVisualization/SuccessionDashboard?publish=yes>.

Ideally, we hope that we met our overarching goal in the eyes of the end user and target audience: the Executive Committee should be able to use these data as easily and directly as possible to make decisions. We designed our dashboard to be intuitive, yet insightful with functionality and interaction between four quadrants. Moving from the upper-left quadrant, clockwise:

1. Quadrant I plots the branch locations for each Branch Manager via a map. Each Branch Manager is represented by a unique shape marker that is consistently presented in other quadrants. When hovering over a marker, additional data are presented such as name and financial metrics. Please observe that each individual is represented by a unique indicator, recognizing their individuality and focusing the Executive Committee on the fact they are making decisions about people – they are not merely a combination of metrics.
2. Quadrant II presents a succession planning grid that plots a Branch Manager's current performance (customer satisfaction score) on the y-axis and overall leadership potential on the x-axis. Minimal thresholds for each axis are set, which creates a four-box grid. For this exercise, greater attention should be paid to those who are both high performing and high potential (i.e., upper-right quadrant), but this matrix could be subsequently used to reassign those who are low performing and low potential or coach and mentor those who are low performing but high potential. Two other sources of information are also plotted: (1) the unique shape marker's (same as in Quadrant I) color represents the Branch Manager's employee turnover rate with a gradient from green (relatively lower turnover) to red (relatively higher turnover); and (2) the linear relationship between leadership potential and current performance. To ease interpretation, no regression statistics are presented, but the statistics are referenced when hovering over the regression line. The predictive model results can be used for future pre-employment selection decisions to project what a candidate's customer satisfaction ratings will be based on his or her leadership potential.
3. Quadrant III compares leadership assessment results across ten indicators of leadership potential (e.g., confidence, forethought, partnership building, business application) for selected Branch Managers. Leadership potential scores are represented by bars, highly successful (Top Performer) and less successful (Contrast Performer) benchmarks are represented by the over-shadowing lines, and the color of the leadership potential bars represent strong (green), moderate (yellow), and weak (red) potential to ease interpretation.



4. Quadrant IV presents peripheral metrics that may add value when making decisions, such as tenure and average quarterly loans, after narrowing the scope to a handful of finalists.

We hope that users intuitively click and drag across a group of high performing and high potential Branch Managers, as depicted in Figure 30.2. In our example, we lassooed three of the more promising candidates: Arnulfo Alequin, Domonique Dismuke, and Mikel Marts in Quadrant II. For these individuals, their data are brought to the forefront, while others' data are suppressed, to focus attention. Notice, other quadrants are updated with their specific information. These two changes leverage movement and color (i.e., shading) to enhance attention, control, and engagement.

Among the three individuals, Domonique has the longest tenure, lowest turnover and highest loan volume, despite being located far outside the main hub of Albuquerque. Although Domonique and Mikel have identical (and strong) customer satisfaction and overall leadership potential, Domonique's profile is more stable, with no leadership indicators below the Contrast Performer benchmark. Mikel's Strength Development, on the contrary, is far below the Contrast Performer benchmark, and may be a contributing factor to his seemingly higher staff turnover – *Does Mikel continually focus on developing the strengths of his staff, or does he routinely point out and try to correct their weaknesses?* It is less important to us that all users arrive at selecting Domonique to be the next District Manager. Instead, we hope to convey that, when executed efficiently, effectively, and interactively, data visualization can foster communication and empower decision-makers to make wiser decisions – notice the ease with which complicated information is displayed and can be digested by decision-makers.

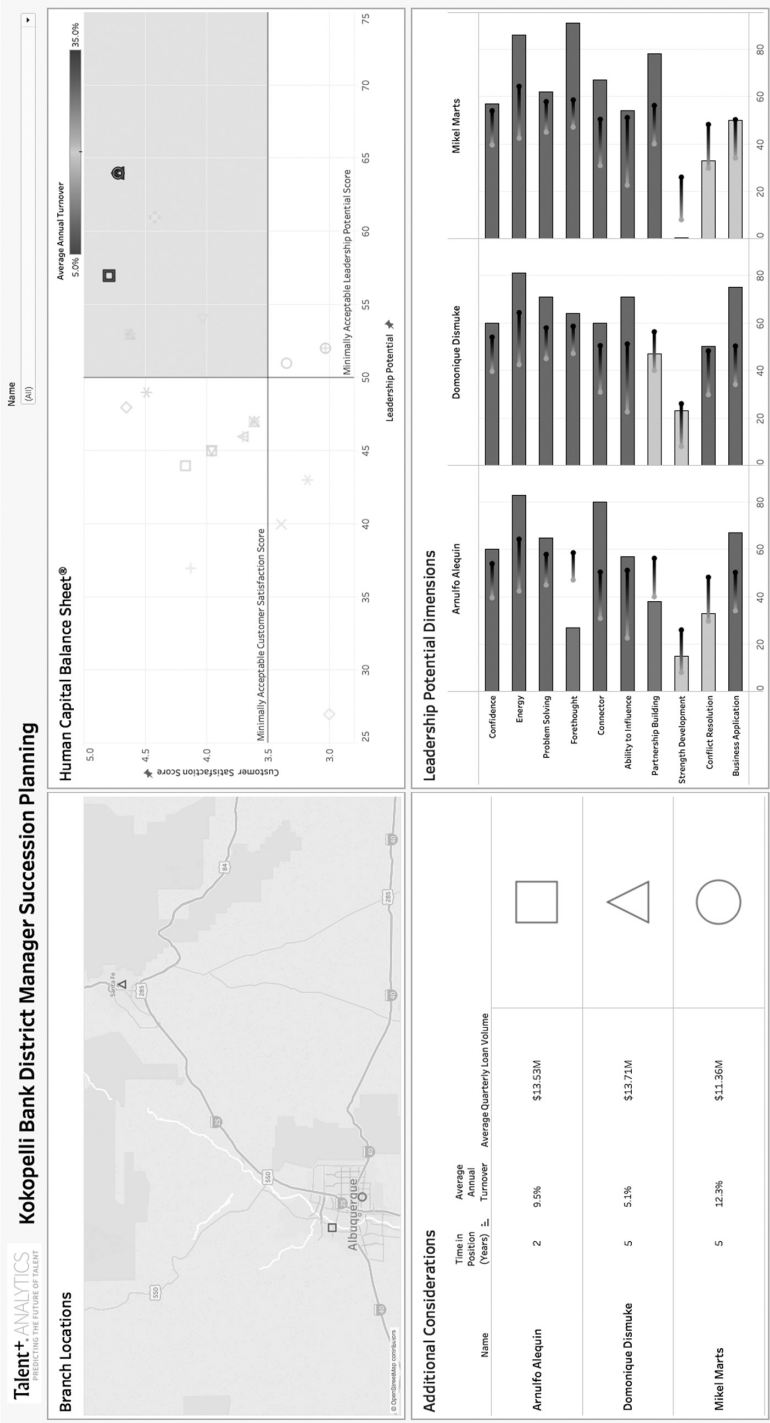
30.4 Visualizing the Bigger Picture

"My inventory goes home every night."

Michael Eisner, Former CEO of the Disney Corporation

An internal or external consultant can design the perfect study, find significant and meaningful results, but ultimately fail if data and analytics are not presented clearly and effectively. Our practical example takes over 500 data points and presents them in an approachable manner that doesn't overwhelm decision-makers with noise, but instead focuses on the few crucial signals. The example makes use of the visual analytic framework and design thinking, progressively narrows the scope of focus, and fosters a sense of ownership, moving decision-makers away from being the traditional, passive audience members and more toward being active participants in the curated discovery process.

Historically, we relied heavily on more traditional presentation tools, such as executive summaries, PowerPoint, and static charts, tables, and graphs. Although carefully analyzed and crafted, static presentations often created more stakeholder



questions than we had answers to in the moment. Leaving a meeting without closure, we would return to our analytics packages to answer these questions, often with a delayed response. As a result, communication suffered and reports and presentations are soon out of sight and mind for busy executives, finding their way to the nearest recycle bin. Interactive, data visualization supports consulting in the moment and, while everyone enjoys a good story, it's even a better story when you get to choose your own adventure. With all data and significant insights in the background, questions can be answered live in the moment with as little as a click or drag. Instead of being labeled "one-hit wonders," data visualization can continue to gain popularity with integrated data feeds and new vistas for exploration and discovery.

While there is substantial literature regarding human perception of graphics and data visualizations (e.g., Cleveland & McGill, 1984; Ware, 2004), there has been a dearth of empirical evidence for its application – the "so what?" Research has weighed heavily on the scientist side of the scientist-practitioner model – understanding visual encoding features such as color, shape, size, gradation, and their combination – but less attention has been paid to the craft or art of visualization. For example, are stories told with visualizations more impactful (e.g., encoding, storage, retrieval, application, deductive and inductive reasoning) than those without visualizations? How do users act on visual information compared to information provided in other mediums? We see so much left to be researched in the realm of data visualization, especially the more recent practitioner innovations such as dynamic, interactive data visualizations. We hope that sharing our experiential knowledge can spur new research directions for academics and may shed light onto the areas that the extant literature does not cover.

Done properly, data visualization has the power to guide human cognition, ultimately leading to better decision-making. Nevertheless, we caution practitioners and consumers to not remove the human element when working with people data or creating visualizations, as subsequent decisions can often impact many lives. Lastly, remember to not simply put "lipstick on a pig." Data visualization is a complement to, but not a replacement for, sound data collection, blending, and analysis. Nevertheless, we look forward to the continuous improvement of data visualization for the mutual benefit of organizations and its many stakeholders.

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